

Introduction to Field Effect Transistors (FETs)

Today's era is a technology-driven generation where we are surrounded by numerous electronic devices all around and now everything is just a click away from us. With the development of technology, we evolved from desktops to supercomputers and in this long journey there is an electronic device that played a very important role in this development, and that is a Transistor. Presently, Field Effect Transistor or FET transistors are broadly utilised transistors, providing the main active component in many integrated circuits. Without these, innovation would be altogether different to what it is presently.

In this article, we are going to have a detailed study on Field-effect transistors, their types, and their applications.

FET Transistor (Field Effect Transistor)



A field-effect transistor (sometimes referred to as a unipolar transistor) is three-terminal active semiconductor equipment where an electric field generated by the input voltage controls the output current. The concept of the field effect semiconductor is based on the idea that a charge on a close-by item can attract charges inside a semiconductor channel. It basically works by utilising an electric field impact and this is how its name originated.

Working of FET Transistor

It has 3 terminals: source, drain, and gate. The working principle is as follows:

- The source and drain terminals are doped with n-type or p-type material to allow electrons or holes to flow between them. The gate terminal is electrically insulated from the channel between source and drain.
- When a voltage is applied to the gate terminal, it creates an electric field in the channel region between source and drain. This electric field modulates the conductivity of the channel, acting like a "valve" that controls the current flow between source and drain.

- For an n-channel FET, applying a positive voltage to the gate attracts free electrons in the channel, increasing its conductivity and allowing more current to flow between source and drain. For a p-channel FET, a negative voltage on the gate repels holes, increasing conductivity.
- By varying the gate voltage, the conductivity and current flow through the channel can be finely controlled. This allows FETs to be used as electronic switches and in amplifying circuits.
- The electric field effect provides a high input impedance and low noise performance, making FETs useful in RF applications and as sensitive analog signal amplifiers.

FET Transistor Types

FET transistors are subcategories into

1. Junction Field-effect Transistor (JFET)
2. Metal Oxide Semiconductor Field-effect Transistor (MOSFET)

Differences between MOSFET and JFET Transistor

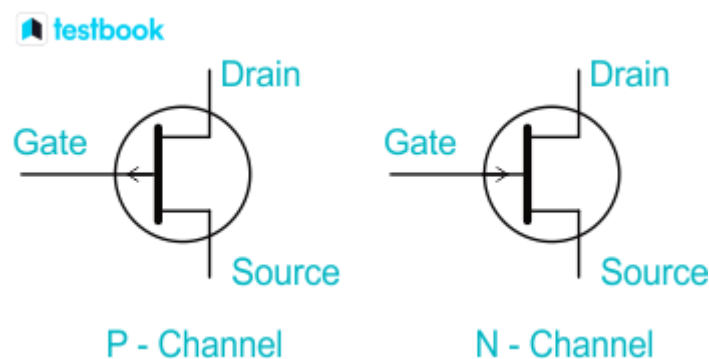
Following are the differences between MOSFET and JFET transistors.

Parameters	MOSFET Transistor	JFET Transistor
Number of Terminal	4 terminal semiconductor device	3 terminal semiconductor device
Mode of Operation	Operates in dual mode that is in both depletion and enhancement modes.	It can only operate in one mode and that is depletion mode.
Order of input impedance	It has a very high input impedance of the order of $10^{14}\Omega$.	It has a high input impedance of order $10^9\Omega$.
Susceptible to Damage	These are more susceptible to damage due to the metal oxide layer which reduces input capacitance.	These are less susceptible to damage due to high input capacitance.
Manufacturing	Manufacturing process is complex	Manufacturing process is simple
Types of Noise Application	These are suitable for high noise applications.	These are ideal for low noise application

Junction Field-Effect Transistor

One of the three terminals on a JFET transistor can control the flow of current among two other terminals. Drain (D), Source (S), and Gate are the three JFET transistor terminals (G). In this case, the gate-source voltage can regulate the current flowing between D and S.

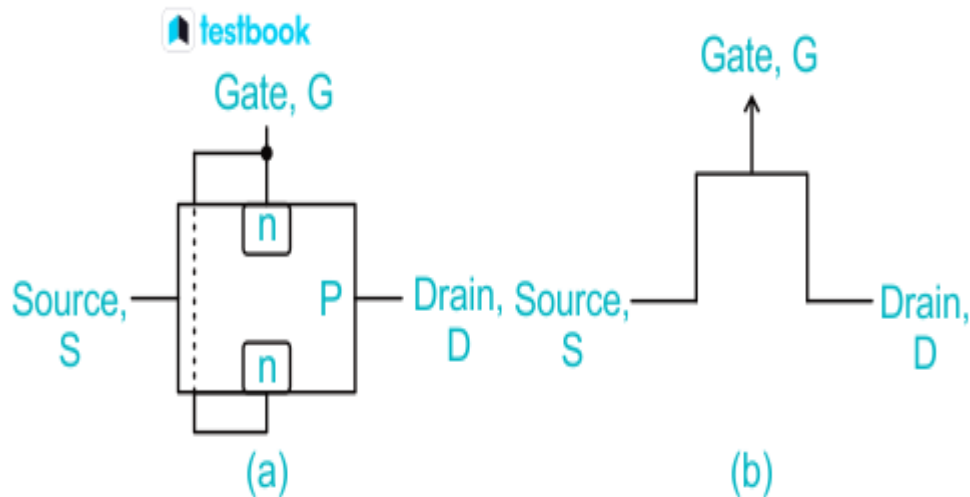
With precise level voltage-operated controllers in analog circuits, JFET is a crucial component. JFETs can be used as switches, voltage-controlled resistors, or to create amplifiers. To replace the BJTs, it also comes in an energy efficient variant. JFETs have relatively low power dissipations and low power transistors also reduce dissipations, which increases the circuit's overall efficiency. It also has a significant benefit over BJTs in terms of input impedance, which is very high.



Types of JFET

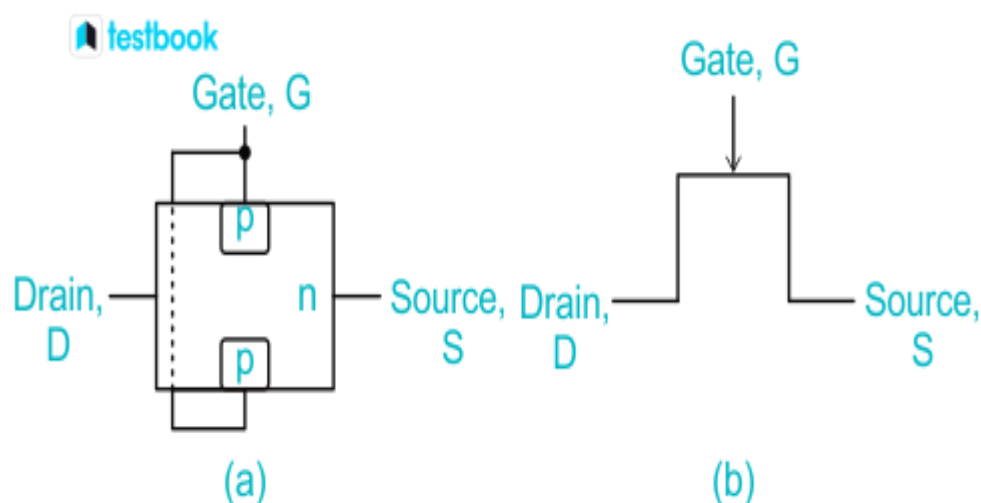
JFET is classified into two types based on the source of current flow. They are N-channel JFET and P-channel JFET.

N-channel JFET



The above figure displays the circuit symbol and schematic for an N-channel JFET. It is evident from the layered structure of the above figure that the N-channel JFET is mostly composed of N-type semiconductors. The source and drain terminals of this bulk substance are on opposite sides. Additionally, it can be seen that this substrate contains two rather tiny P-regions that are internally connected to form the gate terminal. As a result, the gate is of the P-type in this instance whereas the source and drain terminals are of the N-type. This causes two PN junctions to form inside the device, whose analysis reveals the JFET's operating mode.

P-channel JFET



With a few exceptions, the P-channel JFET displays a working mode that is comparable to that of its counterpart, the N-channel JFET. When it comes to P-channel JFETs, the two tiny N-type areas are embedded within the larger P-type

component of the device. As a result, it features an N-type gate terminal and a P-type source and drain, resulting in a P-type channel where the bulk of the charge carriers are holes. Additionally, unlike N-channel JFETs, its circuit symbol's arrow orientation points outwards.

Construction of a JFET

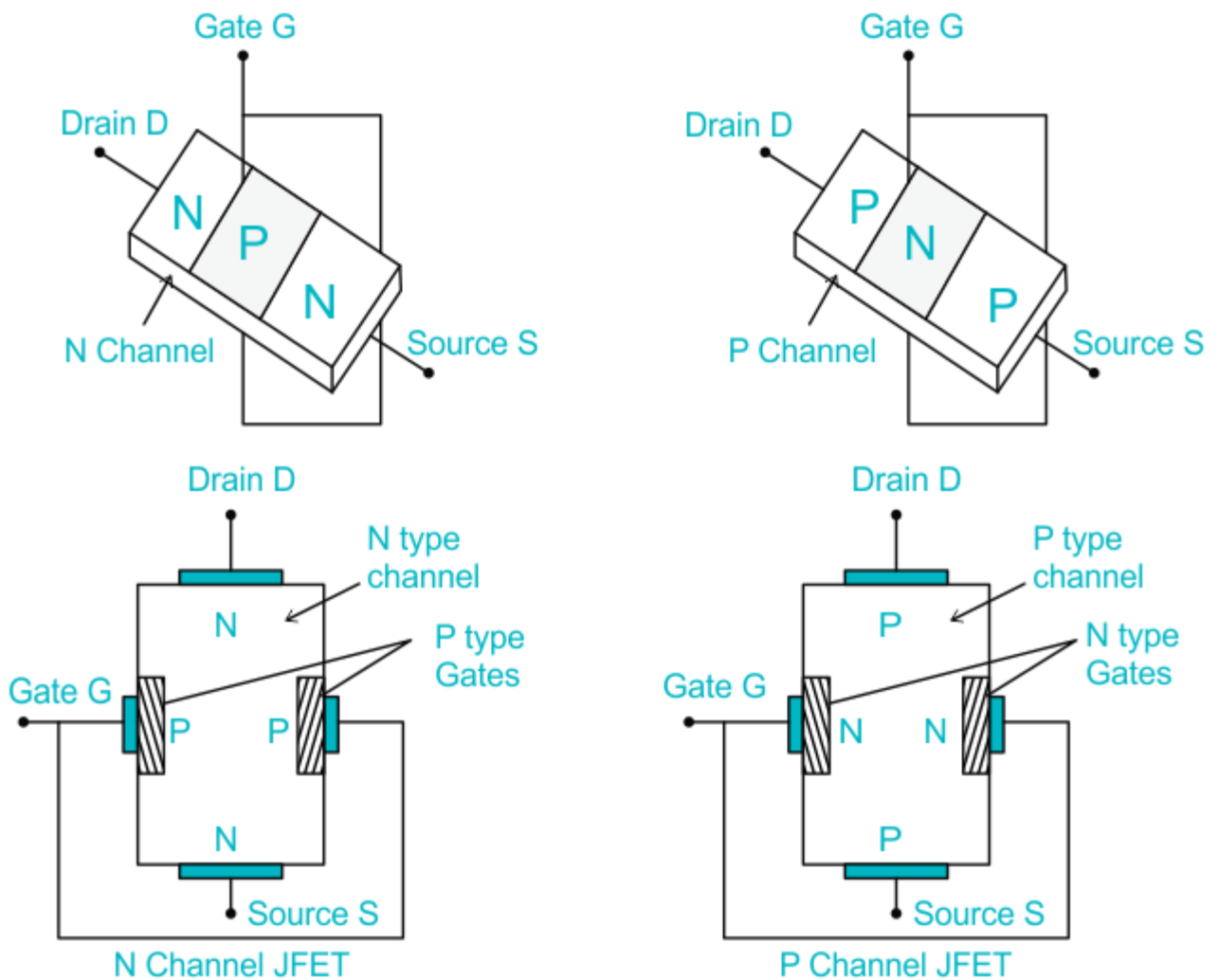
Either N-channel or P-channel is typically used in its fabrication.

The design is quite straightforward. Two tiny bits of P-type silicon material are diffused on opposing sides of an N-type silicon bar, or the channel, in an N-channel JFET to create P-N junctions.

A common terminal known as the gate terminal is brought out after the two P-N junctions that make up diodes or gates are internally joined.

Between its two terminals, D and S, the silicon bar functions as a resistor. The base of a typical transistor is equivalent to the gate terminal.

Two PN-junctions (diodes) called gates are created by these junctions. Channel refers to the space between these gates. The conductor between the source and drain is called a channel. A single lead, known as the gate terminal, is brought out once the two P-regions are internally coupled, as illustrated in the picture below.



The “G” stands for the gate bar, which has resistance-like behavior. At the ends of the bar, ohmic contacts, or direct electric circuits, are formed. The source terminal, which is the lower end of the lead-filled bar, is identified by the letter “S”. Drain terminal, indicated by the letter “D,” is the upper end of a bar that contains lead. It is a device with three terminals: source, drain, and gate. When defining them,

Source: It serves as the terminal via which the majority of carriers enter the bar and move toward the drain (electrons for N-channel and holes for P-channel). The source terminal S is the terminal via which the majority of carriers enter the channel.

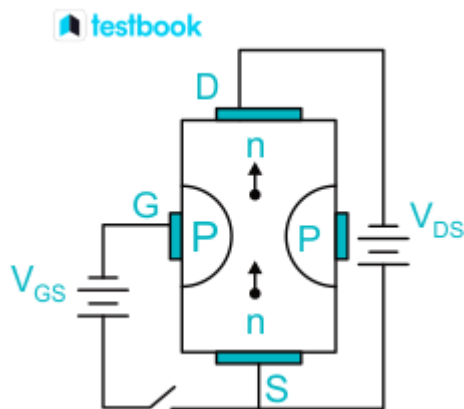
Drain: The drain terminal D is where the bulk of carriers exit the channel, and I_D is the name given to the conventional current that exits the channel at D. This is the exit point for the vast majority of customers.

Channel: The bulk of carriers travel from the source to the drain through this area, which is referred to as the channel and is located between both the source and drain and the two gates.

Gate: There are two P-N junctions produced by alloying, diffusion, or any other method that is accessible, with two internally connected strongly doped impurity regions. Gate G refers to these impurity locations.

Working Principle of Junction Field-Effect Transistor

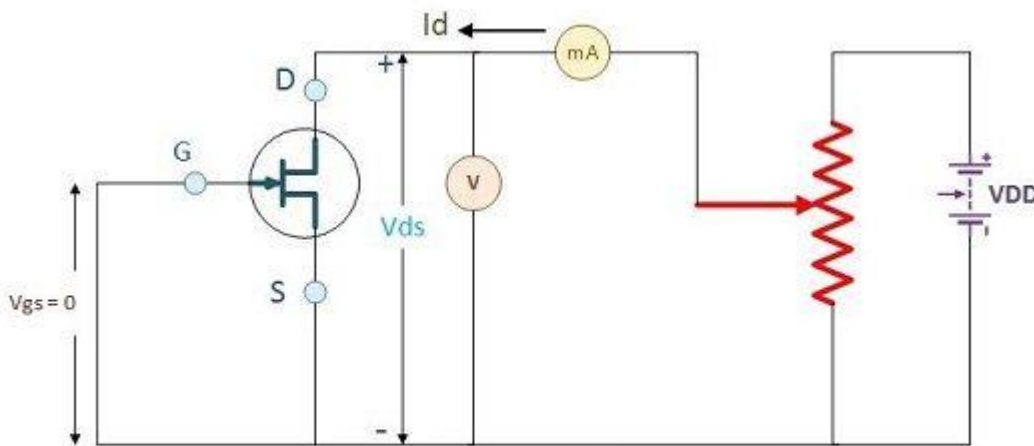
Those two P-N junctions at the sides create depletion layers when voltage V_{DS} is supplied between both the source and the drain terminals and the gate terminal voltage is zero. Through channels between both the depletion layers, electrons go from source to drain. The channel width and subsequently the current conduction through the bar are determined by the width of these depletion layers.



The width of the depletion layers increases whenever a reverse voltage V_{GS} is placed between the gate and source terminals, which reduces the width of a conduction channel and raises the resistance of the conduction channel. As a result, less current flows from source to drain. On the other hand, the breadth of the depletion layer similarly decreases as the reverse voltage V_{GS} is reduced. As a result, the conduction channel widens, becoming a source of the current drain. As a result, the applied voltage on the gate can control the current flowing from source to drain. It is called a field-effect transistor because of this.

Output Characteristics or Drain Characteristics

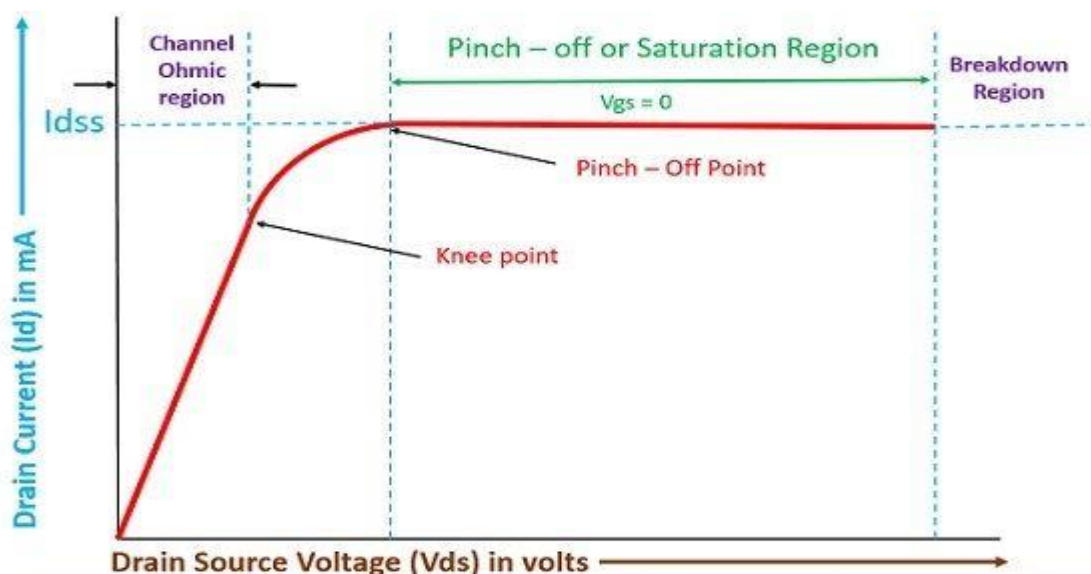
1. **In the absence of external bias:** In this case, as there is no voltage between gate and source terminal, thus, the drain current will flow from drain terminal to source terminal. We have already discussed in the working of JFET that majority charge carriers flow from source to drain and as a consequence of which the current flows from drain to source.



Circuit Diagram for determining drain characteristics with shorted gate for an N-channel JFET

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Now, what does this mean? It clearly implies that the channel width is more as the width of depletion layer will not vary initially because there is no external reverse biasing. This allows a large magnitude of current to flow through the channel.



JFET Output Characteristics with Shorted gate

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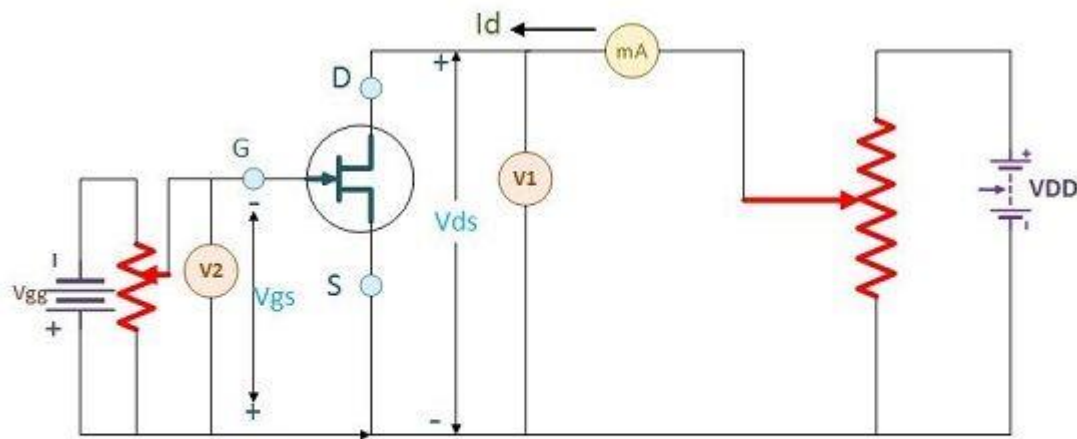
In this case, the N-type channel will simply behave as resistance region. The flow of current from drain to source will create the voltage drop between gate and source. This will eventually result in reverse biasing of the gate-source terminal. The reverse biasing will be more towards drain region than source region.

Terminologies involved in JFET characteristics

1. **Knee Point:** There exists a point in the characteristics curve where the variation of drain current with drain-source voltage appears to be linear. But after this point, the linearity changes into a curve.
2. **Channel Ohmic Region:** The region to the left of the knee point in the characteristics curve is the channel ohmic region.
3. **Pinch-off point:** The point in the curve above which the drain current does not increase further no matter how much we are increasing the drain to source voltage, this point is termed as the pinch-off point.
4. **Pinch-off Voltage:** The voltage at the pinch-off point is termed as pinch-off voltage because at this voltage the current is completely turned to be constant.
5. **Drain-Source Saturation Current:** The drain to source saturation current is the current which becomes constant or completely enters a saturation state.

The region after the pinch-off point in the curve is termed as saturation region. When the JFET is allowed to work as an amplification device, the JFET utilizes this region for operation.

2. **With external bias:** When the external bias is applied to the gate-source terminal, the gate-source terminal becomes reversed bias externally. Obviously, if we are supplying an external voltage, then we can achieve the pinch-off point quite early as compared to the circuit which is not biased.

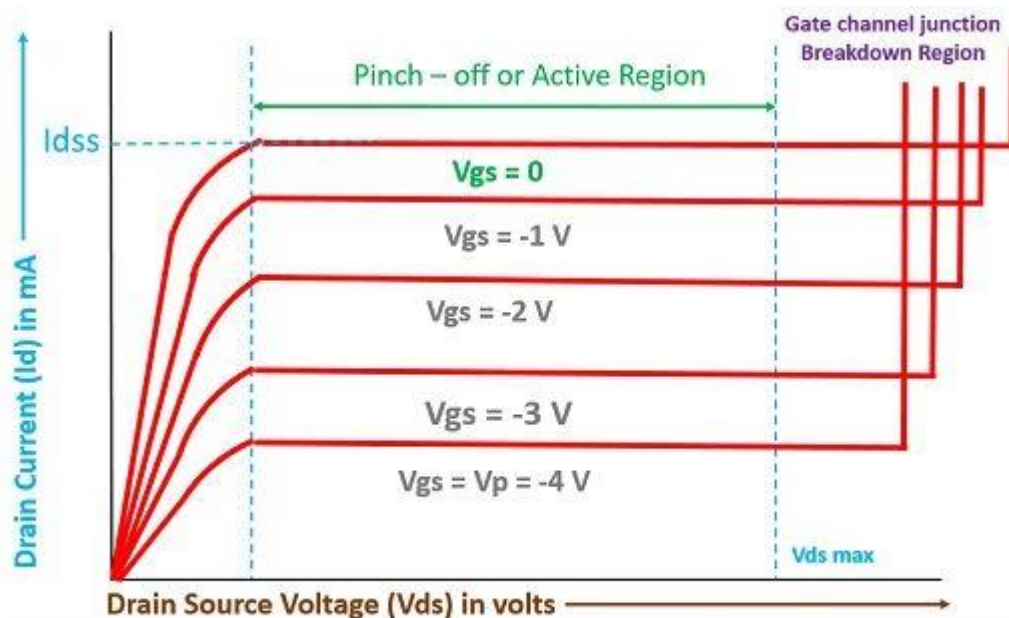


Circuit Diagram for determining drain characteristics with external bias for an N-channel JFET

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It is clearly evident from the characteristics curve of external bias. The different values of voltage give different values of current.

It is to be remembered here that when we are observing the drain characteristics with respect to the variation in drain-source voltage, then the value of gate-source voltage should be kept constant.



JFET Output Characteristics with External Bias

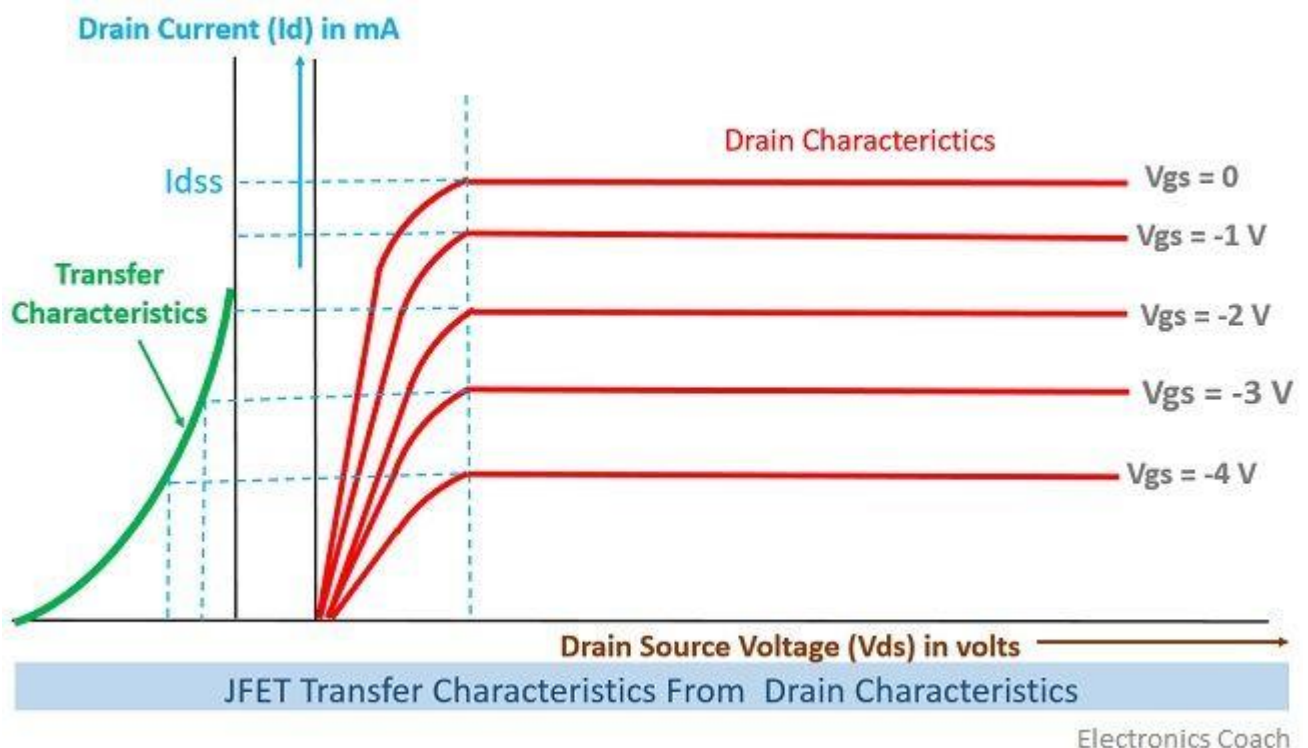
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Transfer Characteristics

The transfer characteristics can be determined by observing different values of drain current with variation in gate-source voltage provided that the drain-source voltage should be constant. The transfer characteristics are just opposite to drain characteristics.

One just needs to remember the concept that in drain characteristics we are keeping the gate-source voltage constant and determining the values of drain current at different values of drain-source voltage while in transfer characteristics we are keeping the value of drain-source voltage constant.

The characteristic curve of transfer characteristics of JFET is described below; it can be easily observed that the value of drain current varies inversely with respect to gate-source voltage when the drain-source voltage is constant.



In this article, we have discussed the output characteristics or drain characteristics and transfer characteristics of an N-channel JFET. What if we want the characteristics of P-channel JFET? Is there will be variation in P-channel JFET characteristics?

The answer to this question is NO. The characteristics curve will remain same. The only difference which will occur is that in case of P-channel the current contributing carriers will be hole rather than electrons in N-channel. Besides, the polarities of gate-source voltage and drain-source voltage will also be reversed in case of P-channel JFET.

Advantages of JFET

A JFET is a voltage controlled, constant current device in which variation in input voltage control the output current. Some of the advantages of JFET are:

1. It has a very high input impedance. This permits high degree of isolation between the input and output circuits.
2. The operation of a JFET depends upon the bulk material current carriers that do not cross junctions. Therefore, the inherent noise of tubes and those of transistors are not present in a JFET.
3. A JFET has a negative temperature co-efficient of resistance. This avoids the risk of thermal runaway.
4. A JFET has a very high power gain. This eliminates the necessity of using driver stages.
5. A JFET has a smaller size, longer life and high efficiency

Applications of Junction Field-Effect Transistor

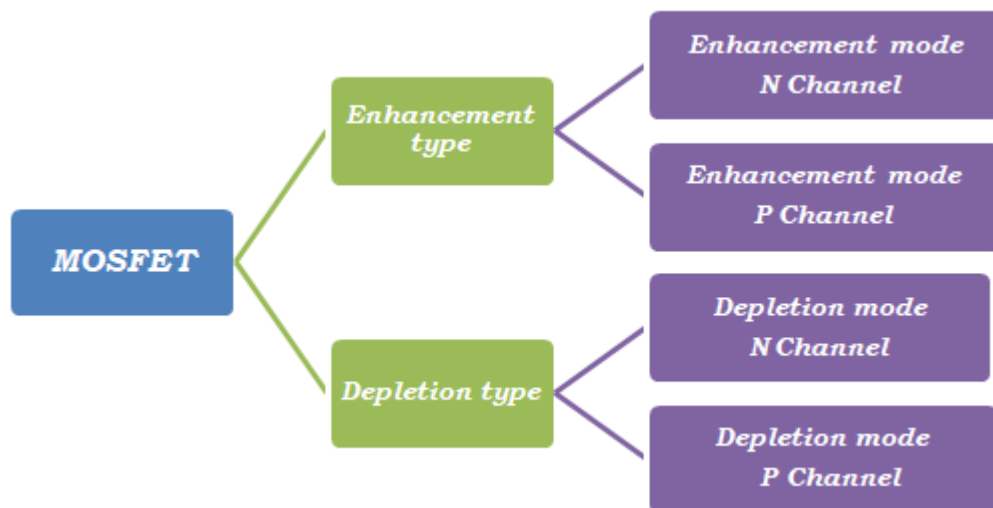
The following list of crucial JFET applications includes

- JFET serves as a source of continuous current.
- JFETs are employed as buffer amplifiers.
- An electronic switch is made with a JFET.
- As a phase-shift oscillator, the JFET is employed.
- The JFET is employed as a wide-band, high-impedance amplifier.
- The JFET serves as a voltage development resistor (VDR) or voltage variable resistor (VVR).
- A chopper is employed with the JFET.
- High Input Impedance Amplifiers use it.

MOSFET

What is MOSFET?

MOSFET stands for Metal Oxide Semiconductor Field Effect Transistor. It is a type of Field Effect Transistor and it is voltage controlled device. It is also called as Insulated Gate Field Effect Transistor (IGFET). It is used for switching or amplifying electronic signals in the electronic devices. It is the most commonly used transistor and it can be used in both analog and digital circuits.

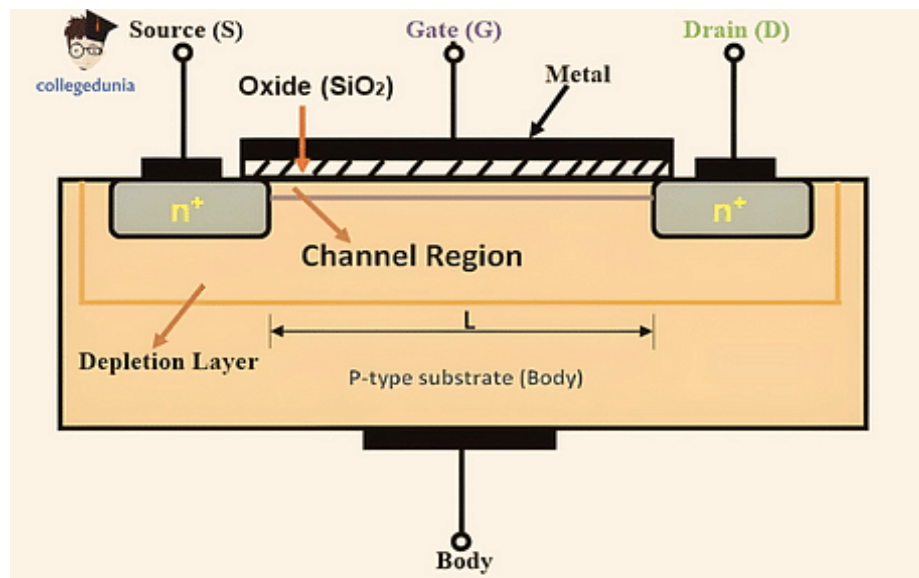


Construction of MOSFET

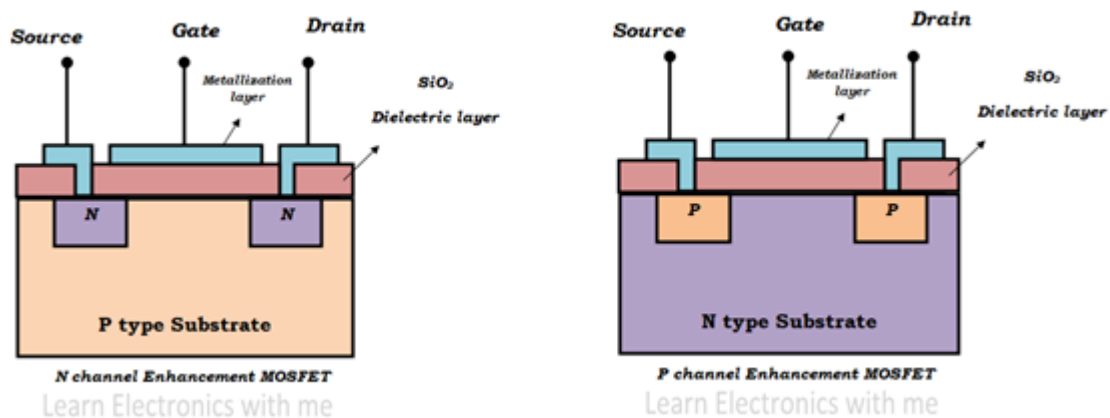
A MOSFET can be constructed as follows:

- The **body of a MOSFET** is typically made from a **p-type semiconductor**.
- The **source and drain terminals** are formed from heavily doped n-type regions on either side of the body.
- These doped regions are often denoted by n^+ in diagrams to indicate their high doping concentration.
- A **layer of silicon dioxide** is deposited on the silicon substrate to provide electrical isolation.
- A thin metallic layer is deposited on **top of the silicon dioxide**, forming a **capacitor-like structure**.
- This metallic layer is then patterned and etched away to define the gate terminal.

- A **voltage source** is then connected between these two n-type areas to construct a DC circuit.

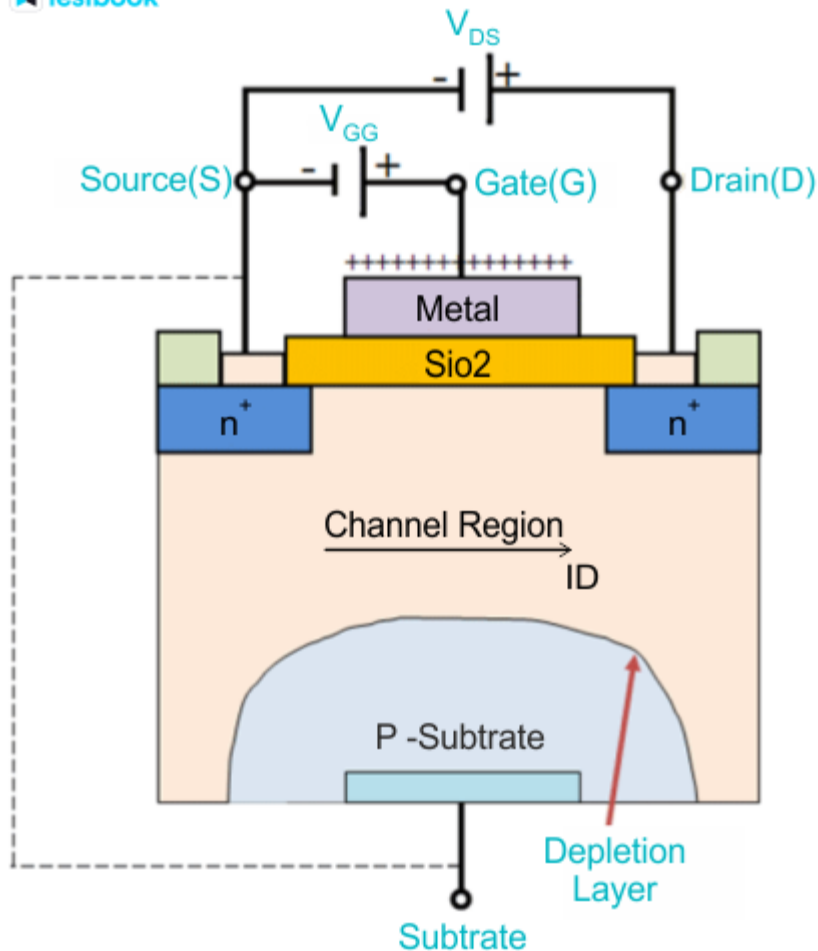


MOSFET circuit diagram



Working Principle of MOSFET

The main working principle of a MOSFET is to control the voltage and the current which is flowing between the source terminal and the drain terminals. We can also say that its work is almost similar to that of a switch.



The semiconductor surface which is present below the oxide layer placed between the source and drain terminal can be reversed from a P-type to an N-type by applying either a positive or negative gate of voltages respectively. When a repulsive force is applied for the positive gate voltage, then the holes present beneath the oxide layer get pushed downward with the substrate present there.

The depletion region of a MOSFET is inhabited by the bounded negative charges which are linked with an acceptor atom. When electrons reach it a channel gets developed. The positive voltage also attracts electrons present in the n^+ source and drain regions towards the channel.

So, when a voltage is applied between the drain and source, the current starts to flow freely between the source and the drain. The flow of gate voltage controls the electrons present in the channel. But, if a negative voltage is applied to it then a hole channel will start to form under the oxide layer of the MOSFET.

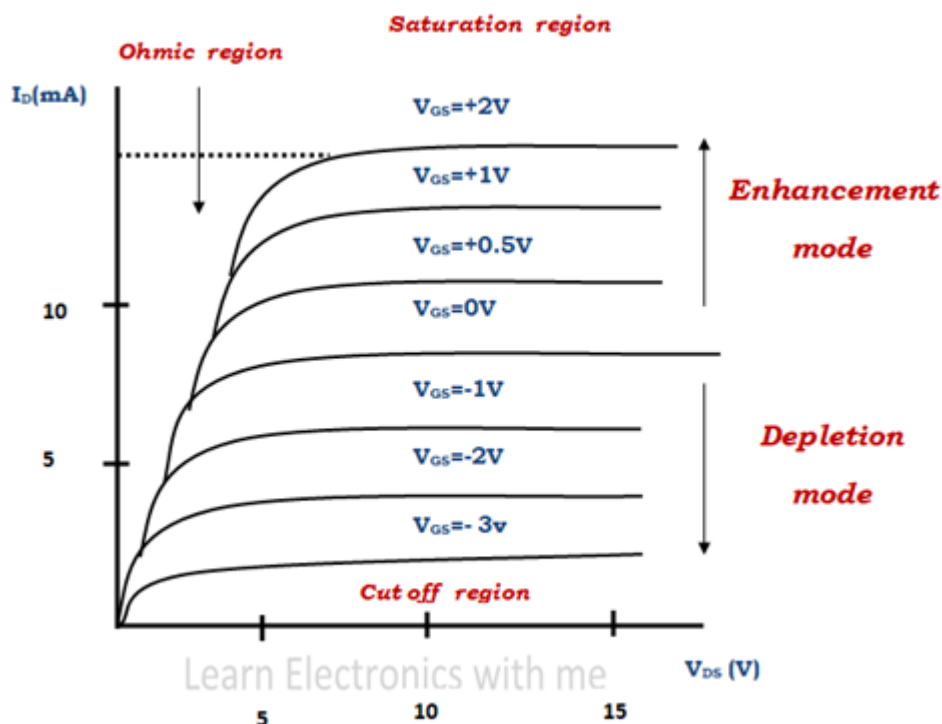
P-Channel MOSFET

A P-channel MOSFET is a type of MOSFET in which the semiconductor material is doped with acceptor impurities, creating an excess of positive charge carriers, or holes. When a negative voltage is applied to the gate, it repels holes from the region beneath the gate, forming a channel of positive charge carriers. This channel allows current to flow from the drain to the source.

N-Channel MOSFET

An N-channel MOSFET is a type of MOSFET in which the semiconductor material is doped with donor impurities, creating an excess of electrons. When a positive voltage is applied to the gate, it attracts electrons to the region beneath the gate, forming a channel of negative charge carriers. This channel allows current to flow from the source to the drain.

V-I Characteristics of MOSFET:



Characteristics of MOSFET

MOSFETs have majorly two characteristics:-

1. Drain characteristics
2. Transfer characteristics

Enhancement Type MOSFET

1. Drain characteristics of Enhancement Type MOSFET

Characteristics between output current and output voltage. O/P i $\rightarrow$ I_D , O/P v $\rightarrow$ V_{DS} , control variable = V_{GS} . Here we will plot a graph between I_D and V_{DS} for various levels of V_{GS} .

Case 1: $V_{GS1} > V_T$

(Here V_T is the threshold voltage)

$$V_{eff} = V_{GS1} - V_T$$

- will directly affect the width of the n-channel
- significant amount of drain current will flow through it.

When V_{DS} is increased to a certain level, drain current becomes constant. This is called pinch-off condition.

$$V_{DS} + V_{GD} - V_{GS} = 0$$

$$V_{GD} = V_{GS} - V_{DS}$$

$$V_{GD} = V_T, \text{ pinch off voltage}$$

Case 2: $V_{GS1} > V_{GS2}$

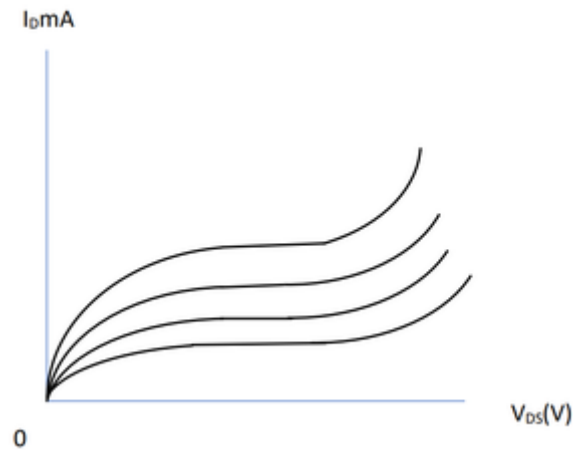
conductivity of 1 $>$ conductivity of 2

$$R_2 > R_1$$

slope of 2 $<$ slope of 1

Now there are few regions in this graph:-

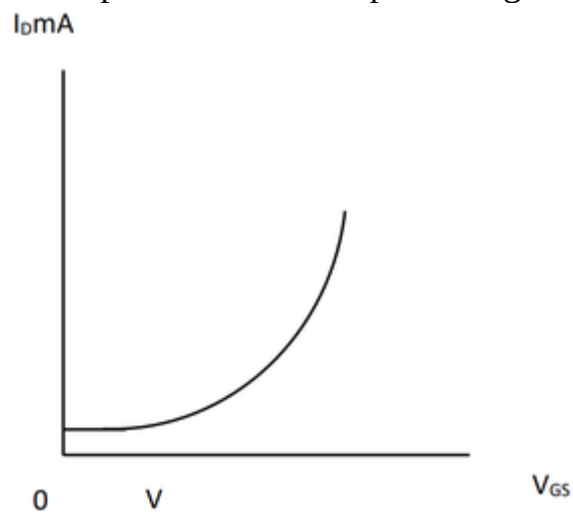
1. **Saturation region** : Region in which drain current is constant.
2. **Triode region** : Left area of locus of V_{DS} saturation.
3. **Cut-off region** : When it can't achieve the threshold voltage, MOSFET remains OFF. Current is 0 amp.



With increasing voltage, current flowing through the terminals increases with voltage. You can see this in the graph. V_{GS} more than the threshold voltage is the condition of flowing current.

2. Transfer Characteristics of Enhancement Type MOSFET

Characteristics between output current and input voltage.

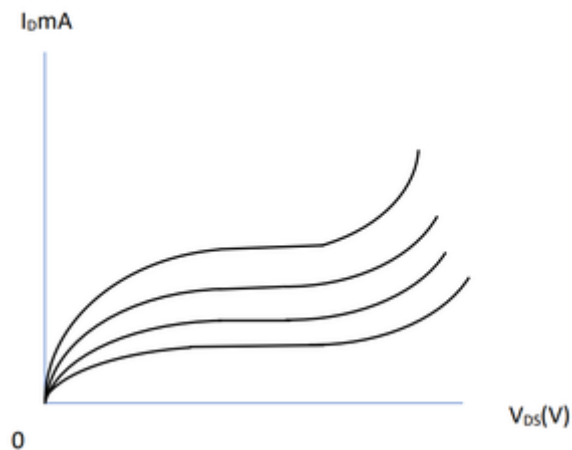


$V = V_T$ (Threshold Voltage)

- When the value of voltage across gate and source is less than the threshold voltage

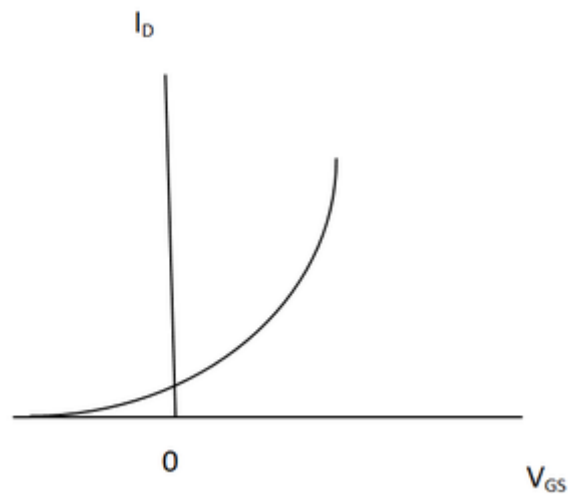
Depletion Type MOSFET

Drain characteristics : I_D V/S V_{DS} for various V_{GS} .



Drain characteristics

Transfer Characteristics : I_D V/S V_{GS} for fixed V_{DS} .



Transfer characteristics

Characteristics of MOSFET

The characteristics of a MOSFET are mentioned as follows:

- It is a tri-terminal
- It is unipolar
- It is controlled by voltage
- It is a high input impedance device

MOSFET Operating Regions

Three functioning areas can be seen in a MOSFET:

Cut-Off Region

The cut-off region is a region where **no conduction occurs**, and so the MOSFET is turned off. MOSFET functions like an **open switch** in this situation.

Ohmic Region

The ohmic region is defined as a **region where the current (I_{DS}) rises as the value of V_{DS} rises**. MOSFETs are used as **amplifiers** when they are designed to function in this range.

Saturation Region

The I_{DS} current of MOSFET remains constant despite an increase in V_{DS} in the saturation area, which occurs when V_{DS} reaches the pinch-off voltage V_P . In this case, **the device will operate as a closed switch**, allowing a saturated value of I_{DS} to pass through it. As a result, whenever MOSFETs are required to perform switching operations, this operating area is picked.

MOSFET as a Switch

MOSFETs can be used as switches to control the flow of current. When the gate voltage is zero, the channel is not formed, and no current flows between the source and drain. When a positive voltage is applied to the gate of an N-channel MOSFET or a negative voltage is applied to the gate of a P-channel MOSFET, the channel is formed, and current flows between the source and drain.

MOSFET as an Amplifier

MOSFETs can also be used as amplifiers to increase the strength of a signal. When a small voltage is applied to the gate, it can produce a large change in current between the source and drain. This property makes MOSFETs very useful in electronic circuits.

Advantages and Disadvantages of MOSFET

The advantages of a MOSFET are mentioned as follows:

- It is used to work at a greater efficiency but with a lower voltage value.
- As it has no gate current so due to this it gives a high input impedance which produces a high switching speed.

The disadvantages of a MOSFET are mentioned as follows:

- It is susceptible to damage due to its thin oxide layer which can get damaged because of electrostatic charges.
- It gets unstable when a voltage gets overloaded.

Disadvantages

Few of the disadvantages are :

- When these devices are functioned at overload voltage levels, it creates instability of the device
- As because the devices have a thin oxide layer, this may create damage to the device when stimulated by the electrostatic charges

Applications

The applications of MOSFET are

- Its amplifiers are widely used in **radiofrequency applications**.
- The MOSFET is a **passive circuit** component.
- DC motors can be regulated using power MOSFETs.
- The **chopper circuit** is designed using MOSFETs.
- It is used as a **switch** to operate as an analog switching device.
- The auto **intensity control** of street lights is done using MOSFETs.

MOSFET can also be used in multiple ways. and those are

- MOSFET as a switch for LED
- MOSFET as a switch for Arduino
- MOSFET switch for ac load
- MOSFET switch for dc motor
- MOSFET switch for negative voltage
- MOSFET as a switch with Arduino
- MOSFET as a switch with a microcontroller
- MOSFET switch with hysteresis
- MOSFET as switch diode and active resistor
- MOSFET as a switch equation
- MOSFET switch for airsoft
- MOSFET as switch gate resistor
- MOSFET as a switching solenoid
- MOSFET switch using an optocoupler
- MOSFET switch with hysteresis

Difference between BJT and MOSFET

The difference between BJT and MOSFET is mentioned in the given table below:

MOSFET	BJT
1) They are mainly of two types named as ○ N-type, and ○ P-type.	1) They are also of mainly two types that are named as ○ NPN, and ○ PNP.
2) It is a device controlled by voltage.	2) It is a device controlled by current.
3) Its input resistance is high.	3) Its input resistance is low.
4) It is used in high-current applications.	4) It is used in low-current applications.
The input resistance of MOSFET is high.	BJTs have a low input resistance.
It is a voltage-controlled device.	A bipolar junction transistor (BJT) is a current-controlled device.
It has a positive temperature coefficient.	It has a negative temperature coefficient.
MOSFET is a unipolar device.	BJT is a bipolar device.
MOSFET is a four-terminal semiconductor device.	BJT is a three-terminal semiconductor device.